ELE 3125 - Signals & Systems

Lecture 8 - Part 1

Properties of the Continuous-Time Fourier Series

$x(t)$ → periodic signal with period T & fundamental frequency ω_0.

Fourier series coefficients denoted by a_k → $x(t) \overset{FS}{\longleftrightarrow} a_k$ [pairing of a periodic signal with its Fourier series coefficients]

Linearity

$$\left.\begin{array}{l} x(t) \overset{FS}{\longleftrightarrow} a_k \\ y(t) \overset{FS}{\longleftrightarrow} b_k \end{array}\right\} \rightarrow z(t) = Ax(t) + By(t) \overset{FS}{\longleftrightarrow} \boxed{c_k = Aa_k + Bb_k}$$

Time-shifting

$$x(t) \overset{FS}{\longleftrightarrow} a_k \rightarrow \boxed{x(t-t_0) \overset{FS}{\longleftrightarrow} e^{-jk\omega_0 t_0} a_k}$$

Time reversal

$$x(t) \overset{FS}{\longleftrightarrow} a_k \rightarrow \boxed{x(-t) \overset{FS}{\longleftrightarrow} a_{-k}}$$

If $x(t)$ is even: $a_{-k} = a_k$

If $x(t)$ is odd: $a_{-k} = -a_k$

Time-scaling

$$x(t) = \sum_{k=-\infty}^{\infty} a_k e^{jk\omega_0 t} \rightarrow \boxed{x(\alpha t) = \sum_{k=-\infty}^{\infty} a_k e^{jk(\alpha\omega_0)t}}$$

Multiplication

$$\left.\begin{array}{l} x(t) \overset{FS}{\longleftrightarrow} a_k \\ y(t) \overset{FS}{\longleftrightarrow} b_k \end{array}\right\} \rightarrow x(t)y(t) \overset{FS}{\longleftrightarrow} \boxed{h_k = \sum_{l=-\infty}^{\infty} a_l b_{k-l}}$$

Conjugate and Conjugate Symmetry

$$x(t) \overset{FS}{\longleftrightarrow} a_k \rightarrow \boxed{x^*(t) \overset{FS}{\longleftrightarrow} a_{-k}^*}$$

If $x(t)$ real → $x(t) = x^*(t)$ → $\boxed{a_{-k} = a_k^*}$ → $|a_{-k}| = |a_k|$

- $x(t)$ real & even → $a_k = a_{-k}$ → $a_{-k} = a_k^*$, $a_k = a_k^*$ → FS coefficients real & even
- $x(t)$ real & odd → FS real & odd

Parseval's Relation for Continuous-Time Periodic Signals

$$\boxed{\frac{1}{T}\int_T |x(t)|^2 dt = \sum_{k=-\infty}^{\infty} |a_k|^2}$$

$x(t) = a_k e^{jk\omega_0 t}$

$|x(t)|^2 = |a_k e^{jk\omega_0 t}|^2 = |a_k|^2 |e^{jk\omega_0 t}|^2$

$e^{jk\omega_0 t} = \cos(k\omega_0 t) + j\sin(k\omega_0 t)$

$|e^{jk\omega_0 t}| = |\cos(k\omega_0 t) + j\sin(k\omega_0 t)| = \sqrt{\cos^2(k\omega_0 t) + \sin^2(k\omega_0 t)} = 1 \Rightarrow \boxed{|e^{jk\omega_0 t}|^2 = 1} \Rightarrow \boxed{|x(t)|^2 = |a_k|^2}$

$\frac{1}{T}\int_T |a_k|^2 dt = \frac{1}{T}|a_k|^2 \int_0^T dt = \frac{1}{T}|a_k|^2 [t]_0^T = \frac{1}{T}|a_k|^2 [T-0] = \frac{1}{T}|a_k|^2 (T) = |a_k|^2$

$\Rightarrow \boxed{\frac{1}{T}\int_T |a_k|^2 dt = |a_k|^2}$ $|a_k|^2$: average power in the k-th harmonic component

Parseval's relation:

Total average power in a periodic signal = sum of average powers in all of its harmonic components